

Suraksha

On-device scam defence for first-time digital-banking users.

Trilingual (Gujarati / Hindi / English) - works offline - explains every verdict.

THE MAVERICK EFFECT AI CHALLENGE 2026

Financial Safety for Rural India

THE PROBLEM

A first-time user, alone with a scammer

Rural, first-time digital-banking users face scams built to rush and frighten them:

- Scam calls and 'digital arrest' threats
- Fake UPI collect-requests ('approve to receive money')
- Phishing links and lookalike bank sites
- Advance-fee 'loans' and lottery/KBC bait

They need help in their own language, on the phone in their hand, right now.

THE INSIGHT

Scams mutate. The psychology doesn't.

Keyword models rot: numbers, brands and schemes change every week.

What never changes is the manipulation grammar - urgency, authority, fear, secrecy, reward, the OTP ask.

So we model the tactics, not the wording. That is durable, explainable, and language-portable.

WHAT IT DOES

Check -> Explain -> Act

- CHECK: paste or speak a message, call, UPI request or link
- EXPLAIN: instant verdict, highlighted phrases, tactic radar, archetype, and what they'll ask next
- ACT: guided playbook with the 1930 helpline and a ready-to-file complaint
- LEARN: adaptive 'Scam or Safe?' drills that train on your own mistakes

Every verdict is read aloud, in Gujarati, Hindi or English.

THE DIFFERENTIATOR

A radar of what is being done to you

8 tactic axes scored 0-100: Urgency, Authority, Fear, Reward, Secrecy, Credential ask, Fake trust, Irreversibility.

Matched to 16 known Indian fraud archetypes.

The demo moment: not 'SCAM 87%' but 'fear + authority + secrecy - this is a digital-arrest fraud, and next they will ask you to stay on the call and transfer money to clear your name.'

ARCHITECTURE

A calibrated hybrid ensemble

- Neural: multilingual transformer + logistic head, in-browser (Transformers.js)
- Rules: pure-TypeScript, instant, 100% explainable, 8 groups
- Tactics: 8-axis scorer + archetype matcher

Fusion: 0.45 neural + 0.40 rules + 0.15 tactics; hard overrides floor OTP/PIN and APK links to DANGER.

HONEST METRICS

Small numbers, honestly reported

- 5-fold CV F1 mean 0.844
- Ablation: rules-only F1 0.786 -> full ensemble 0.923
- Precision 0.786 -> 1.000 while holding recall (fewer false alarms, no more misses)
- Threshold tuned for recall ≥ 0.95

Corpus being expanded to a trilingual ~1,200-row target with native speakers.

ON-DEVICE & OFFLINE

No server. No data plan. No PII.

- All inference runs in the browser - nothing sent for analysis
- ~30MB model downloaded once, then cached; installable PWA
- Works fully offline (demo it in aeroplane mode)

Privacy by design is also the reason it costs zero to scale.

SCALABILITY & IMPACT

Zero marginal cost, compounding reach

- Device compute -> scales to millions on a free tier
- Portable core -> website widget, SMS/IVR fallback, WhatsApp helper (documented paths)
- Adaptive drills + shareable family-poster -> each user protects their elders

Impact spreads faster than cost.

ETHICS & WHAT'S NEXT

Assistant, not a guarantee

- We never claim to tap a live call or fake bank integration
- Safety-first defaults; every helpline and URL verified and cited
- Next: finish the trilingual corpus, native-review all copy, add /history & /report

Open the live URL on your phone, switch to Gujarati, and try a real scam.

MIT-licensed, public repo, reproducible notebook.